

Toy Example

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Simulation Setup

We consider four data-generating mechanisms. In all cases, we generate

$$A_i \sim \text{Bernoulli}(0.5), \quad \mathbf{X}_i \sim \mathcal{N}(\mathbf{0}, I_5), \quad \mathbf{W}_i \sim \mathcal{N}(\mathbf{0}, I_5)$$

Let $\beta_X, \beta_W \in \mathbb{R}^5$ be unit-norm vectors, and define

$$\eta_i = \frac{\mathbf{X}_i^\top \beta_X + \mathbf{W}_i^\top \beta_W}{\sqrt{2}}, \quad \alpha_0 = -0.5, \quad \alpha_1 = 0.5.$$

(i) Ideal case. We define

$$S_i = \text{expit}(\alpha_{A_i} + \eta_i), \quad Y_i | S_i \sim \text{Bernoulli}(S_i).$$

In this case,

$$S_i = \mathbb{E}[Y_i | \mathbf{X}_i, \mathbf{W}_i, A_i], \quad \mathbb{E}[Y_i | S_i, A_i, \mathbf{W}_i] = S_i,$$

so S_i is a perfectly calibrated risk score.

(ii) S misspecified but $\mathbb{E}[Y|S, A]$ lies within the beta-calibration family and $\mathbb{E}[Y | S, A] = \mathbb{E}[Y | S, A, W]$. We define

$$S_i = \eta_i, \quad Y_i | S_i, A_i \sim \text{Bernoulli}(\text{expit}(\alpha_{A_i} + S_i)).$$

Here,

$$\mathbb{E}[Y_i | S_i, A_i, \mathbf{W}_i] = \mathbb{E}[Y_i | S_i, A_i],$$

so S_i is sufficient for predicting Y_i , but is not calibrated.

(iii) S misspecified and $\mathbb{E}[Y|S, A]$ doesnt lie within the beta-calibration family and $\mathbb{E}[Y | S, A] = \mathbb{E}[Y | S, A, W]$. We define

$$S_i = \eta_i, \quad Y_i | S_i, A_i \sim \text{Bernoulli}(\text{expit}(\alpha_{A_i} + \gamma(S_i + 0.5S_i^2))),$$

with $\gamma = 1$. In this case,

$$\mathbb{E}[Y_i | S_i, A_i, \mathbf{W}_i] = \mathbb{E}[Y_i | S_i, A_i],$$

but the regression function is nonlinear in S_i , so beta calibration are misspecified.

(iv) Additional information in $\mathbf{W}$. We define

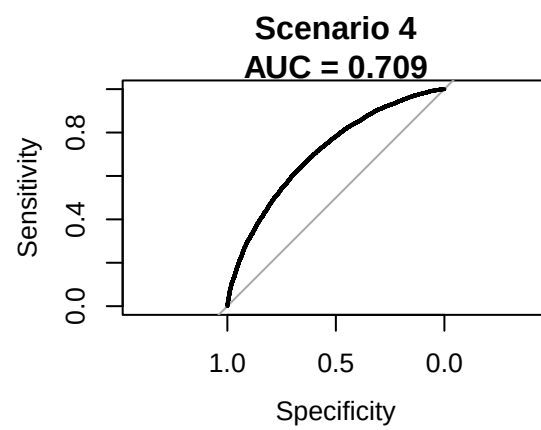
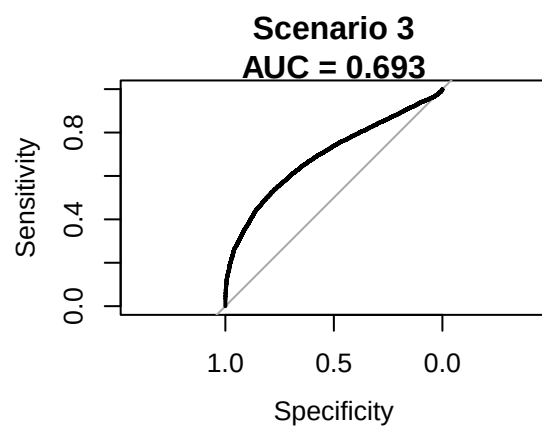
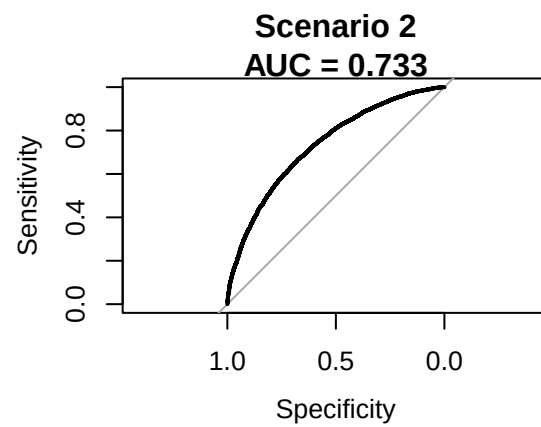
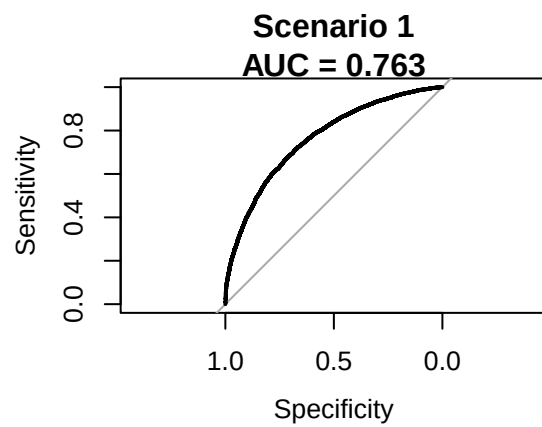
$$S_i = \mathbf{X}_i^\top \beta_X, \quad Y_i | S_i, \mathbf{W}_i, A_i \sim \text{Bernoulli}(\text{expit}(\alpha_{A_i} + S_i + \mathbf{W}_i^\top \beta_W)).$$

Here,

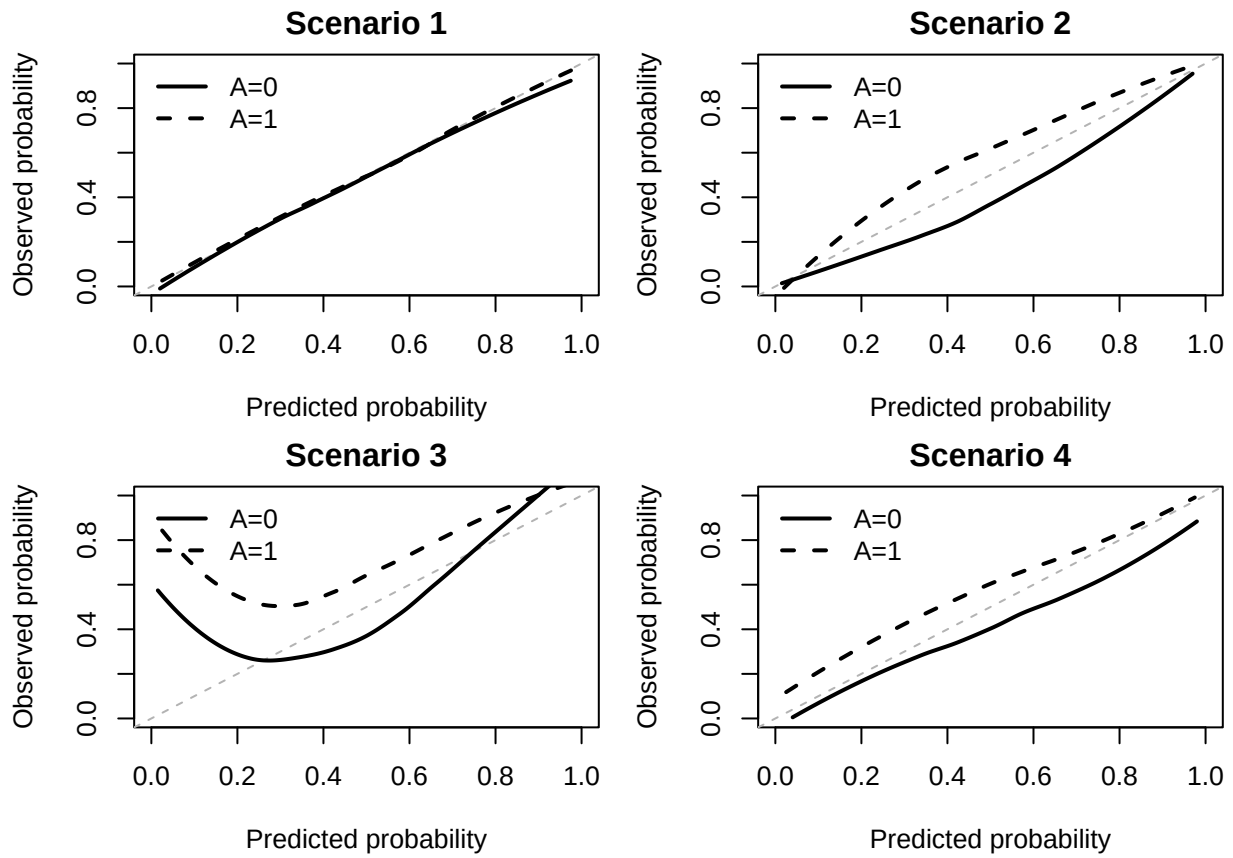
$$\mathbb{E}[Y_i | S_i, A_i, \mathbf{W}_i] \neq \mathbb{E}[Y_i | S_i, A_i],$$

so $\mathbf{W}_i$ provides additional predictive information beyond S_i .

ROC Check

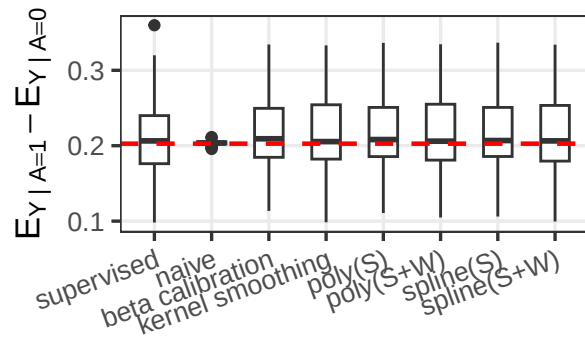


Calibration Check

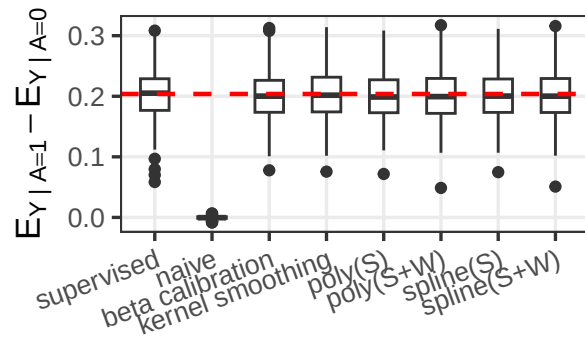


Point Estimate

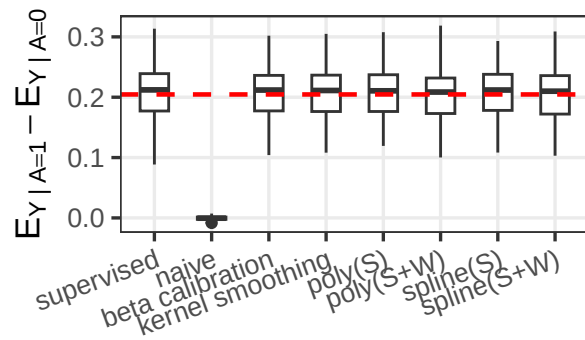
Scenario 1



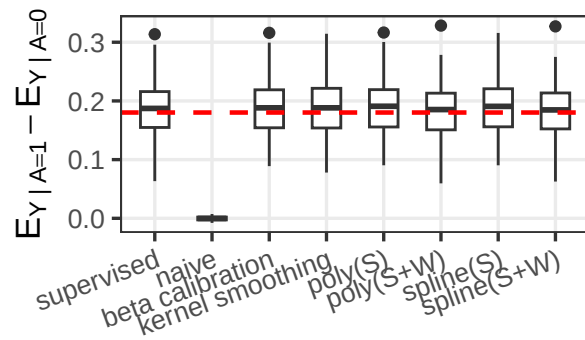
Scenario 2



Scenario 3

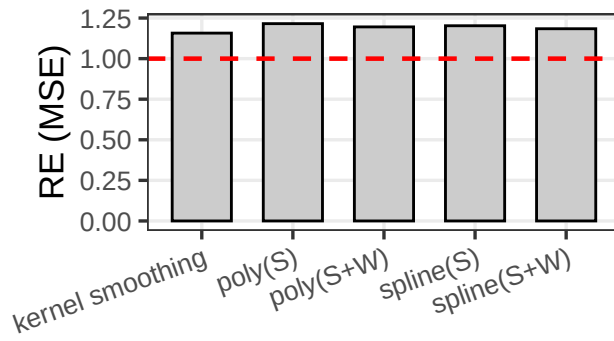


Scenario 4

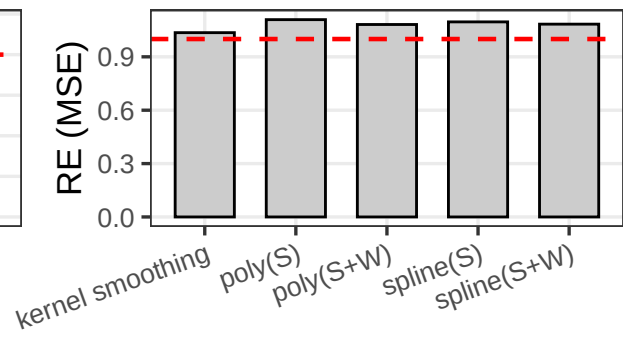


Relative Efficiency

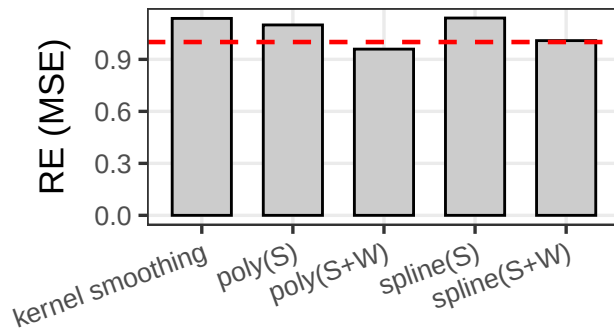
Scenario 1



Scenario 2



Scenario 3



Scenario 4

